

Shrabya Bhattarai

Hattiesburg, MS | +1 (601) 307 4516 | shrabya.com.np | github.com/source-shrabya | linkedin.com/in/shrabya-bhattarai

EDUCATION

University of Southern Mississippi

Bachelor of Science in Computer Engineering

Hattiesburg, MS

Aug 2024 – May 2028

- GPA: 3.91/4.0 | Relevant Coursework: Digital Electronics, Computer Networks, Operating Systems, Embedded Systems, Data Structures

TECHNICAL SKILLS

Languages: Python, C, C++, SQL

Data & Analytics: Python scripting, SQL, data cleaning, CRM databases, automated reporting

AI & Vision: YOLO, vision-language models, model fine-tuning

Hardware: Arduino, sensors, motor drivers, Mecanum drive

Tools & Frameworks: Flask, Git

EXPERIENCE

4th Dimension Marketing Solutions

Cincinnati, OH

Data Analyst Intern

May 2026 – Aug 2026

- Write Python scripts that automate the collection and organization of daily field sales data into structured databases, replacing manual entry across multiple sales locations.
- Clean and validate raw sales records from physical marketing campaigns, resolving inconsistencies before the data enters reporting and analysis workflows.
- Build automated reporting pipelines that generate weekly sales performance summaries for the marketing team, reducing turnaround on reporting requests.
- Analyze multi-location sales data to identify performance trends and surface insights used to adjust campaign strategy.
- Maintain a centralized CRM database using SQL to track leads, sales figures, and campaign outcomes across the team.

TechMatrix

Kathmandu, Nepal

Junior Software Engineer Intern

May 2023 – Aug 2024

- Worked on the IoT side of the team's software projects, writing parts of the backend in Python and Flask.
- Collaborated with a small team of engineers on shared projects.

PROJECTS

Wildfire Smoke Detection on Edge Hardware

Python, YOLO, Vision-Language Models

Research Project | University of Southern Mississippi

Jan 2026 – Present

- Co-developing a two-stage system: a lightweight image classifier screens every camera frame, and a small vision-language model reviews only the frames it is unsure about.
- Put the paper together, found and reviewed the reference literature, and helped train the models.
- Currently helping debug the models; paper in progress.

Robo Soccer — Autonomous Arduino Robot

Arduino, C++

Finalist | Google Developer Group (GDG) USM Competition

Oct 2025

- Designed and built a custom Arduino-powered robot for autonomous soccer play.
- Programmed movement algorithms for ball tracking and precision control.
- Reached the finals among multiple university teams.

Line-Following Robot Car

Arduino, C++, Sensors, Motor Driver

Personal Project

- Built and programmed a robot car from scratch that follows a black line.
- Modified a Mecanum wheel design for better maneuverability and stability.
- Tuned sensor readings and motor response for smooth, accurate path tracking.

ACTIVITIES & LEADERSHIP

USM Open Source Community (OSC)

University of Southern Mississippi

Social Media Lead & Co-Founder

March 2026 – Present

- Co-founded the OSC club and manage the @oscusm social media presence, growing the community to 200+ student members through workshops, open-source contribution drives, and event coordination.